MIL-H-24606A(SH)
6 December 1985
SUPERSEDING
MIL-H-24606(SH)
21 May 1982
(See 6.7)

MILITARY SPECIFICATION

HOSE ASSEMBLIES, CHLOROSULFONATED POLYETHYLENE IMPREGNATED, DOUBLE SYNTHETIC JACKET, WITH COUPLINGS, FIRE FIGHTING AND OTHER WATER SERVICE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers chlorosulfonated polyethylene impregnated, heavy duty, double synthetic jacket, fire fighting and water service hose.

1.2 Classification.

1.2.1 Military part number. The Military part number for the hose assemblies covered by this specification shall consist of the letter "M", the basic number of the specification and a coded dash number (see 1.2.2) as shown in the following example:

M24606 – 250

Basic number of specification
Hose assembly nominal size

1.2.2 Hose assembly size. The hose assembly size (nominal inside diameter) is identified by three digits (see table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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TABLE I. Hose assembly size.

Code no.	Assembly size (inches)
150	1-1/2
175	1-3/4
250	2-1/2
300	3
350	3-1/2
400	4

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

WW-C-621 - Coupling Assembly, Hose, (Fire, Woven Jacketed, Rubber or Cambric-lined or Unlined Linen).

MILITARY

MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-H28/10 - American National Hose Coupling and Fire-Hose Coupling Threads.
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.
FED-STD-191 - Textile Test Methods.
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 276 - Standard Methods for Identification of Fibers in Textiles. (DoD adopted)
D 380 - Standard Methods of Testing Rubber Hose. (DoD adopted)
D 412 - Standard Test Methods for Rubber Properties In Tension. (DoD adopted)
D 413 - Standard Test Methods for Rubber Property-Adhesion to Flexible Substrate. (DoD adopted)
D 518 - Standard Test Method for Rubber Deterioration - Surface Cracking. (DoD adopted)
D 573 - Standard Test Method for Rubber - Deterioration in an Air Oven. (DoD adopted)
D 1059 - Standard Test Method for Yarn Number Based on Short-Length Specimens.
D 1149 - Standard Test Method for Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen). (DoD adopted)
D 1907 - Standard Test Method for Yarn Number by the Skein Method.
D 2240 - Standard Test Method for Rubber Property - Durometer Hardness. (DoD adopted)
D 2256 - Standard Test Method for Breaking Load (Strength) and Elongation of Yarn by the Single-Strand Method.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection and tests (see 4.3 and 6.6).

3.2 Material. Material shall be new and unused. The hose shall have been manufactured not more than 1 year prior to date of delivery.

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3.2.1 Rubber. The rubber used in the liner shall be compounded from natural, synthetic, or a mixture of the two stocks.

3.2.2 Jacket yarns. The yarns (see 4.6.10) used in both jackets shall be of virgin synthetic yarns as specified below:

Filler yarn: A high strength, low stretch filament polyester.
Warp yarn: A high strength, low stretch filament polyamide nylon 6,6 stable high bulked yarn comprising an entangled convoluted mass of continuous filaments, the bulked yarn as supplied for weaving having a denier at least 8 percent greater than the denier of the unbulked yarn. The bulk in the yarn shall be retained during the weaving process such that the denier of the bulked yarn in the woven jacket is at least 6 percent greater than the denier of the unbulked yarn and capable of accepting total saturation of impregnating compound.

Yarns that do not accept total saturation of impregnating compound are not acceptable.

3.2.2.1 Abrasion - taber method. The number of abrasion cycles required to completely abrade through one or more filler yarns of the outer jacket for all sizes hose shall be not less than 2800 cycles.

3.2.2.2 Abrasion - Factory Mutual Research Corporation method. The number of abrasions cycles required to completely abrade through the outer jacket for all sizes hose shall be not less than 8000 cycles.

3.2.3 Impregnating compound. Impregnating compound shall be chlorosulfonated polyethylene, bright orange in color.

3.2.4 The contractor shall provide a certificate of compliance certifying that materials used were those specified and that no substitutions were made (see 6.2.2).

3.3 Construction. The hose shall consist of a seamless liner of rubber bonded to the inner woven jacket over which the impregnated outer woven jacket is placed. Unpressurized hose shall assume the shape of a flattened oval to facilitate storage.

3.3.1 Liner. The liner shall be a single ply, synthetic rubber extrusion of uniform thickness. The liner shall be free from dirt, blisters and other imperfections. Inside surface shall be smooth and free from corrugations. Liner thickness shall be as specified in table II when measured in accordance with 4.6.2. Liner properties before and after aging shall be as specified in table III.

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TABLE II. Liner thickness.

Hose size	Min thickness (inches)	Max thickness (inches)
150	0.040	0.052
175	.040	.052
250	.040	.052
300	.049	.062
350	.049	.062
400	.058	.070

TABLE III. Liner properties.

Property	Value
Shore hardness, A scale	50 ± 5
Tensile strength, before aging	1600 lb/in^2 minimum
Tensile strength, after aging	70 percent retention after aging
Elongation, before aging	400 percent minimum
Elongation, after aging	70 percent retention after aging
Ozone resistance	No cracks after 100 hours exposure

3.3.1.1 Adhesion. The adhesion between the backing and the jacket, between the backing and the liner and between the liner and the jacket shall be such that the rate of separation of a 1-1/2 half inch strip of lining, transversely cut shall be not greater than 1 inch per minute under a 12 pound weight.

3.3.2 Jackets. The jackets shall be seamless and have the fillers woven around the tube throughout its length. At no point shall the tube show through either jacket nor shall the filler yarn protrude through the warp yarn. Jacket construction shall minimize water absorption. Yarn tensile strengths and filler pick count shall be as specified in table IV.

TABLE IV. Yarn tensiles.

Size	Tensile strength (pounds min)				Filler pick count/inches min	
	Warp		Filler			
	Inner jacket	Outer jacket	Inner jacket	Outer jacket	Inner jacket	Outer jacket
150	30	41	145	145	7-1/2	7-1/2
175	30	41	145	145	7-1/2	7-1/2
250	41	53	185	145	7-1/4	8
300	53	53	205	205	7-1/4	7-1/2
350	53	53	248	248	7-1/4	7-1/2
400	53	53	287	287	6-3/4	6-3/4

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3.3.3 Impregnation. The outer jacket only shall be impregnated. Impregnation shall be a total heat set infusion of chlorosulfonated polyethylene. Impregnation or coating types cured with natural evaporation are not acceptable. Impregnation shall be prior to the jackets being combined.

3.3.3.1 Total saturation shall consist of total immersion of the outside jacket into a liquid compound of chlorosulfonated polyethylene and shall include a method to force the compound to all surfaces in the jacket.

3.3.4 Inside diameter. The inside diameter of the hose shall be not less than the trade size of the hose as specified in table I, except that the 2-1/2 inch size shall have an inside diameter of not less than 2-9/16 inches.

3.3.5 Length and weight. Unless otherwise specified (see 6.2.1), the fire hose lengths and weights shall be furnished as specified in table V.

TABLE V. Length and weight of hose assembly.

Size	Average length per lot feet (min)	Before sampling each length feet (min)	After sampling each length feet (min)	Weight of coupled length pounds (max)
150	50	48	47	20
175				21-1/2
250				35
300				46-1/2
350				62
400				72

3.3.6 Couplings and gaskets. Unless otherwise specified (see 6.2.1), each length of hose shall be fitted with a set (one male and one female) of cast or forged brass couplings, and gaskets, conforming to type A, style 1 of WW-C-621. Unless otherwise specified (see 6.2.1), the coupling threads shall be as specified in table VI, FED-STD-H28 and FED-STD-H28/10. Gaskets shall be inserted at face of expansion ring in the coupling bowl.

TABLE VI. Coupling threads.

Size	Thread designation
150	1-1/2 - 11-1/2 NPSH
175	1-3/4 - 11-1/2 NPSH
250	2-1/2 - 7-1/2 NH
300	3 - 6 NH
350	3-1/2 - 6 NH
400	4 - 6 NH (SPL)

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3.4 Performance.

3.4.1 Hydrostatic proof pressure. The hose, complete with couplings, shall not leak, sweat, or show leakage at the couplings, and shall not show breakage of jacket yarns when it is subjected to the applicable hydrostatic pressure as specified in table VII for a minimum of 15 seconds and tested for elongation, twist, warp and rise in accordance with 4.6.1.1.

TABLE VII. Pressure testing requirements.

Size	Proof pressure (lb/in^2)	Kink pressure (lb/in^2)	Burst pressure (lb/in^2)	
			Inner jacket only	Complete double jacket assembly
150	600	500	585	900
175	600	500	585	900
250	600	500	585	900
300	600	400	585	900
350	600	300	540	800
400	600	300	540	800

3.4.1.1 Hose elongation. When subjected to the proof pressure test in accordance with 4.6.1.1, the hose shall not elongate more than 8 percent for sizes 150 through 250 and not more than 10 percent for size 300 and not more than 13 percent for sizes 350 and 400 when measured from an initial pressure of 10 pounds per square inch (lb/in^2).

3.4.1.2 Hose twist. When subjected to the proof pressure test in accordance with 4.6.1.1, the hose shall not twist more than 29 degrees per foot for sizes 150 and 175 and not more than 12-1/2 degrees per foot for sizes 250 through 400. A maximum twist to the left of 2 degrees per foot shall be permitted while the pressure is being raised; but any final twist shall be to the right.

3.4.1.3 Hose warp. When subjected to the proof pressure test in accordance with 4.6.1.1, the hose shall not deviate more than 20 inches from a straight line drawn from center to center of the fittings at the ends of the hose.

3.4.1.4 Rise. When subjected to the proof pressure test in accordance with 4.6.1.1, the hose shall not rise, or lift off the flat surface upon which it rests.

3.4.1.5 Kink. When tested in accordance with 4.6.1.3, the hose shall not leak, burst or show other signs of failure.

3.4.1.6 Low temperature flexibility. When tested in accordance with 4.6.3, the hose shall show no damage to the jackets or lining, and shall not leak when subjected to the proof pressure of the hydrostatic test upon completion of the low temperature flexibility test.

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3.4.1.7 Water absorption. When tested in accordance with 4.6.4, the maximum weight gain shall not exceed that as specified in table VIII.

TABLE VIII. Water absorption weight gain.

Hose size	Weight gain pounds (max)
150	5
175	5
250	7
300	8-1/2
350	12
400	15

3.4.2 Burst pressure. The hose shall not burst at pressures less than those as specified in table VII for the respective size of hose when tested in accordance with 4.6.1.2. The pressure shall be applied to the hose when it is lying straight and when it is held in a circular arc of 27-inch radius.

3.5 Friction or head loss. The first article sample shall conform to the friction or head loss maximum requirements as specified in table IX when tested in accordance with 4.6.5.

TABLE IX. Maximum allowable friction loss, lb/in^2 per 50 feet.

Hose size	Flow rate gallons per min	Allowable loss per 50 feet (lb/in^2)
150	100	17
175	125	9
250	250	8
300	308	8-1/2
350	298	4
400	300	2-1/2

3.6 Marking. Each hose assembly shall be marked at a point beginning not less than 4 feet from each end with stenciled letters at least 1-inch high in indelible ink of a contrasting color. Marking shall include:

(a) Number of this specification (including revision letter).
(b) Size and length.
(c) Tested to 600 lb/in^2.
(d) Month and year of manufacture.
(e) Manufacturer's name or FSCM code no.

3.7 Labeling of hose. When specified (see 6.2.1), hose shall be approved and labeled by a nationally recognized testing group or agency (see 6.4), adequately equipped to perform testing of fire hose.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection lot. Hose lengths with couplings of the same size with the same construction materials, produced under essentially the same conditions, and offered for inspection at one time shall be considered a lot for purposes of inspection.

4.2.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Unless otherwise specified (see 6.2.1), first article inspection shall be performed on one hose assembly (see 3.1). The inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the contractor's current inventory provided that the unit meets the requirements of the specification and is representative of the design, construction, and manufacturing techniques applicable to the remaining units to be furnished under the contract.

4.3.1 First article test report. The contractor shall prepare first article test reports in accordance with the data ordering document (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

(a) Individual examination and test (see 4.4.1).
(b) Sampling for test (see 4.4.2).

4.4.1 Individual examination and tests. Each length of hose or hose assembly shall be subjected to the following examination and tests:

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(a) Visual examination and dimensional inspection, except for inside diameter (see 4.5 through 4.5.2.1).
(b) Hydrostatic pressure test (see 4.6.1.1).

4.4.2 Sampling for tests.

4.4.2.1 Sampling for burst test (see 3.4.2). A random sample of hose lengths shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-1. From each sample selected, a 3-foot section shall be removed and tested in accordance with the applicable test methods as specified in table X. Tests of the sample selections shall alternate between bursting the inner jacket only sample and the complete double jacket assembly sample. Both of the inner jacket only samples and the complete double jacket assembly samples shall alternate between bursting the straight sample and the curved sample. There shall be no failures.

TABLE X. Test for physical properties.

Property	Requirement paragraph	Method in FED-STD-601, or ASTM	Reference paragraph or footnote
Hydrostatic proof pressure	3.4.1 and table VII	ASTM D 380	4.6.1.1
Burst pressure	3.4.2 and table VII	ASTM D 380	4.6.1.2
Elongation	3.4.1.1	ASTM D 380	
Twist, degrees per foot	3.4.1.2	ASTM D 380	
Warp	3.4.1.3	ASTM D 380	
Rise	3.4.1.4	ASTM D 380	
Kink under pressure	3.4.1.5 and table VII	ASTM D 380	4.6.1.3
Construction	3.3	-	4.6.2
Accelerated aging	3.3.1, table III	ASTM D 573	
Tensile strength of tube	3.3.1, table III	ASTM D 412	1/
Ultimate elongation of tube	3.3.1, table III	ASTM D 412	1/
Hardness of liner	3.3.1, table III	ASTM D 2240	
Thickness of liner	3.3.1, table II	Method 2011	
Adhesion	3.3.1.1	ASTM D 380	4.6.7
Ozone resistance	3.3.1, table III	ASTM D 1149	2/ 4.6.6
Inside diameter	3.3.4	Method 2341	4.5.2.2
Yarn tensile	3.3.2, table IV	ASTM D 2256	
Pick count	3.3.2, table IV	-	4.6.2
Low temperature flexibility	3.4.1.6	-	4.6.3
Water absorption	3.4.1.7	-	4.6.4
Friction head loss	3.5	-	4.6.5

1/ For rubber lining, die A of ASTM D 412 shall be used for cutting the specimen.
2/ Specimen shall be prepared in accordance with ASTM D 518, procedure C.

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4.4.2.2 Sampling for construction, accelerated aging, tensile strength, ultimate elongation, ozone resistance, liner thickness, hardness, adhesion, pick count and inside diameter. Samples for the jacket construction, and for accelerated aging, tensile strength, ultimate elongation, adhesion and for liner thickness, ozone resistance, pick count, hardness and inside diameter of the tube may be taken from samples used in the burst test. In the event that insufficient samples can be obtained from the burst test samples, a random sample of hose lengths shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-1. The AQL shall be 2.5 percent defective. From the hoses sampled, 3-foot sections shall be cut and tested, in accordance with 4.6.2 and 4.6.3.

4.4.2.3 Sampling for water absorption. One 50-foot length each of size 150 and size 250 hose without end couplings shall be subjected to the test as specified in 4.6.4.

4.4.2.4 Sampling of yarn construction. Four packages of each type of yarn used in the construction shall be retained from the yarn lot corresponding to that material used for producing a given hose lot. Three samples shall be used from each of these packages to determine the average minimum yarn tensile strength and percentage of denier increase when compared to the denier of unbulked yarn. Four each inner and outer jacket samples, 60 inches in length, will be retained for use in determining the percentage of denier increase when compared to the denier of unbulked yarn, and for the determination of the warp yarn composition and type. If entangled filament is supplied directly from a fiber producer, certification and yarn test results shall be supplied in lieu of in plant test results on percentage of denier increase when compared to denier of unbulked yarn.

4.4.2.5 Sampling for weight determination. A random sample of coupled hose lengths shall be selected from each lot in accordance with MIL-STD-105, at inspection level S-1 for the hose weight determination specified in 4.5.2.3. The AQL shall be 2.5 percent defective.

4.4.2.6 Sampling for abrasion - taber method. A random sample of hose lengths shall be selected by the contractor prior to offering the lot to the Government for acceptance. This sample shall be 30 inches long, with one sample selected for every five-hundred lengths per lot. A minimum of one sample shall be selected for each lot. In addition, the Government Quality Assurance Representative will select one additional 30-inch sample per 500 lengths per lot prior to acceptance, with a minimum of one sample selected from each lot. The Government reserves the right to perform the abrasion test on additional lengths at the destination for a period of 60 days after receipt of any hose lot.

4.4.2.7 Sampling for abrasion test - Factory Mutual Research Corporation method. A second sampling shall be selected as prescribed in 4.4.2.6.

4.5 Examination.

4.5.1 Visual examination. Each length of hose shall be examined for the defects listed in FED-STD-162. In addition, each hose shall be examined to determine if the color meets the requirements as specified in 3.2.3.

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4.5.2 Dimensions.

4.5.2.1 Length. The distance between the inside edges of the couplings shall be determined as specified in method 2411 of FED-STD-601, except that the measurement shall be made while a hydrostatic pressure of 10 lb/in^2 is applied to the specimen as specified in ASTM D 380. The lengths of all hose in the lot shall be in accordance with 3.3.5. The entire lot shall be rejected if the average length per lot is less the requirement specified in table V, and an individual hose assembly shall be rejected if it is less than the corresponding length specified in table V.

4.5.2.2 Inside diameter. The inside diameter shall be determined as specified in method 2341 of FED-STD-601, except that not less than two measurements shall be made on the unit or test unit. The measurements of hose with couplings shall be made when the 3-foot length is taken for the burst test, or when the 3-foot length is taken for low temperature flexibility testing.

4.5.2.3 Weight determination. The 50-foot samples for the weight determination shall be drained of any residual water and shall then be weighed to the nearest 1/10 pound and the weight recorded.

4.6 Tests. Hose samples selected shall be subjected to the applicable tests as specified in table X.

4.6.1 Hydrostatic pressure tests.

4.6.1.1 Hydrostatic proof pressure. Each length of hose in the lot with couplings attached, or length of hose remaining after removal of portions on which couplings have been attached by the manufacturer for use in other tests, shall be subjected to a hydrostatic pressure test and to the tests specified in table X. The hose assembly shall be held for a minimum of 15 seconds at the test pressure. The pressure applied to the hose length shall be as specified in table VII. Each hose assembly shall be index-marked with a pencil or other suitable device at a point immediately adjacent to the back side of each coupling. No slippage or leakage at the couplings or hose shall occur when the pressure specified in table VII is applied to the length. Individual hose assemblies shall be rejected if they fail to meet the requirements of 3.4.1 and table VII.

4.6.1.2 Burst pressure. Hose assemblies from each lot shall be subjected to the burst pressure specified in table VII. Samples shall alternate between the inner jacket only burst and the double jacket hose assembly burst. The burst pressure test shall be determined as specified in ASTM D 380, except the outer jacket shall be carefully removed without damaging the inner jacket for the inner jacket only burst samples.

4.6.1.3 Kink. One full length of hose from each lot shall be completely filled with water, taking care to remove all the air within the hose. The hose shall be pressurized to not less than 10 lb/in^2 and then kinked 18 inches from the free end by tying the hose back against itself so that there will be a sharp kink. The pressure shall then be raised to that specified in table VII and then immediately released. The entire lot shall be rejected if the sample fails the requirements of 3.4.1.5.

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4.6.2 Construction and physical property tests. The thickness of the lining shall be determined as decribed in method 2011 of FED-STD-601.

Method	Characteristic
5050	Construction (see 3.3 and table IV)

After removal of the specimens used in tests of the lining, the remaining specimens of hose jacket shall be unraveled as necessary and tested for consistency. In addition to the above test, the filler pick count shall be determined and checked in accordance with 3.3.2.

4.6.3 Low temperature flexibility. A 3-foot section of hose shall be exposed to a temperature of -40 ± 2 degrees Fahrenheit (°F) for a period of 24 hours. At the end of the exposure period and while maintained at the (-40 ± 2°F) exposure temperature, the hose shall be rapidly bent 180 degrees double on itself, first one way and then the other. The hose shall then be examined for evidence of cracking or breaking of the jacket and liner. The hose shall then be allowed to return to room temperature over a 24 hour period, after which it shall be subjected to the applicable hydrostatic proof pressure. Leakage shall be cause for rejection.

4.6.4 Water absorption. The 50-foot sample hose shall be conditioned in an oven at 158 ± 5°F for 72 hours, then cooled to room temperature. The sample shall then be weighed to the nearest 1/10 pound and the weight recorded. The sample shall then be submerged in water for 1 hour, removed from the water and allowed to drip dry for 10 minutes by hanging vertically. The sample shall then be weighed again to the nearest 1/10 pound. The dry, or first weight, shall be subtracted from the wet, or second weight, to determine compliance in accordance with 3.4.1.7 and table VIII. The entire lot shall be rejected if the sample fails to meet the requirements of 3.4.1.7 and table VIII.

4.6.5 Friction or head loss. The friction or head loss test shall be made with the coupling gaskets in place. Two or more lengths shall be coupled together. Friction or head loss from hose inlet to outlet shall be measured with calibrated gauges, a manometer, or through the use of a gauge and a calibrated flow meter. The maximum allowable head loss shall be no greater than the values as specified in table IX.

4.6.6 Ozone resistance. Specimens for ozone resistance of the tube shall be prepared in accordance with ASTM D 518 procedure C. Specimens shall be tested in accordance with ASTM D 1149. Specimens shall be elongated 15 percent, ozone concentration 100 ± 5 parts per hundred million by volume. Temperature shall be 100 ± 3°F and exposure time shall be 100 hours.

4.6.7 Adhesion. The adhesion test specimens shall be prepared in accordance with the procedure specified in ASTM D 380, for woven jacketed rubber lined hose. The specimens shall then be tested in accordance with the static mass method specified in ASTM D 413 using a 180 degree peel strip specimen.

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4.6.8 Abrasion.

4.6.8.1 Tabor method. The abrasion test shall be conducted in accordance with FED-STD-191, method 5306. The abrasion machine shall be of the type equipped with a vacuum nozzle to continuously remove dust, lint and any disintegrated or worn cloth from the test specimen. The vacuum level shall be maintained at a setting of 60. The abrasion machine shall be equipped with a pair of H-22 calibrade wheels and shall be conducted with a 1000 gram load. Five specimens shall be cut from the outer jacket of each sample selected in accordance with 4.4.2.6 so that both folds of the jacket shall be included in the abrasion path, and the specimens shall measure 4-1/4-inches square and tested as follows:

(a) The specimens shall be affixed to Taber No. S-36 mounting cards, with the inside portion of the outer jacket fabric in contact with the adhesive side of the cards. A 1/4-inch diameter hole shall be punched into the center of the specimen and card, and a rubber cement shall be applied to the perimeter of the fabric to prevent displacement of the filler yarns during the test. It may be necessary to unravel one or two warp ends on the sides of the specimens before applying the cement to properly anchor the filler yarns.
(b) The card and specimen shall then be placed on the E 100-125 specimen holder, with the outer surface of the fabric facing up. The clamp plate and nut shall then be lightly tightened to anchor it to the specimen holder.
(c) The tester shall be started and the specimens shall be examined at intervals not to exceed one-hundred cycles until the end-point is reached. The end-point shall be that point at which one or more filler yarns is completely abraded through or severed in two, with the number of cycles recorded to the nearest 10 cycles.
(d) The number of cycles to complete abrasion of the filler yarns from the five specimens from each sample shall be averaged and recorded. Also, each individual value of the average shall be recorded.

The entire lot of hose submitted for acceptance at one time shall be rejected if the average number of cycles to completely abrade the filler yarn for any one sample is below the requirement of 3.2.2.1.

4.6.8.2 Factory Mutual Research Corporation method. The 30-inch samples selected in accordance with 4.4.2.7 shall be attached with couplings at each end. A cap shall be placed on one end of the sample and it shall be filled with water at a pressure of 125 lb/in^2 and firmly anchored horizontally. A Norton standard 5 by 2 by 1/2 inches abrasion wheel designation 37C36-K.V.K. or equal, shall be reciprocated back and forth for 8000 cycles along the crown of the hose (see 6.5). The wheel shall be prevented from rotating and shall exert on the hose its full weight plus half of the 1 pound of the moving arm. The stroke of the abrading wheel shall be 14 inches. The frequency of the cycle shall not exceed 30 per minute. The contact surface of the abrasion wheel shall be unused at the start of the test.

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4.6.8.2.1 Examine the samples at 500 cycle intervals until the end point is reached. The end point shall be the total number of cycles until the outer jacket is completely abraded through and the wheel first makes contact with the inner jacket warp yarns or 8000 cycles which ever comes first. The entire lot of hose shall be rejected if the sample fails to meet the requirements of 3.2.2.2.

4.6.9 Determination of warp yarn composition and type. The warp yarn composition and type shall be verified by subjecting the samples from 4.4.2.4 to two test methods. The first test method is contained in ASTM D 276, section 16, "Fiber Identification Using Solubility Relationships". If after conducting this method the results are positive for nylon, the sample shall then be subjected to ASTM D 276, section 26, "Fiber Melting Point". A positive test for the presence of the condensation product of hexamethylenediamine and adipic acid or nylon 6,6 shall be if the fiber specimens melt at 254 ± 2°C. Should any one sample fail to meet this criteria, then the entire lot shall be rejected.

4.6.10 Yarn denier. Three specimens 40 inches in length shall be cut from 60-inch specimens and tested in accordance with ASTM D 1059 for determining the denier of the bulked yarn in the woven jackets. ASTM D 1907, option 2 shall be used for determining the denier of the bulked yarn as supplied for weaving.

4.7 Inspection of packaging. The preservation, packaging, packing and marking of the hose shall be inspected to determine compliance with section 5.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.3.)

5.1 General. When hose is subjected to hydrostatic tests at the manufacturer's plant, the hose shall be carefully drained and the jackets thoroughly dried before preparation for shipment.

5.2 Preservation-packaging and packing. Hose shall be cleaned, preserved-packaged level A or C and packed level A, B or C as specified (see 6.2.1) and marked in accordance with MIL-P-775.

6. NOTES

6.1 Intended use. This hose is intended for use in shipboard applications wherever double cotton jacket hose or rubber covered hose is used where additional abrasion resistance is required such as non-skid decks on shipboard. This hose is not for use in AFFF/PKP hose reels.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Classification (see 1.2).
(c) Whether or not first article is required (see 3.1, 4.3 and 6.6).

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(d) If length is other than specified (see 3.3.5).
(e) If couplings are other than specified (see 3.3.6).
(f) Whether couplings should be male or female (see 3.3.6).
(g) If coupling thread is other than specified (see 3.3.6 and table VI).
(h) When approval and labeling is required (see 3.7).
(i) Number of first article hose assemblies, if other than specified (see 4.3).
(j) Level of preservation, packaging and packing required (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.4	Certificate of compliance	DI-E-2121	---
4.3.1	First article inspection report	DI-T-4902	10.1.b

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.4 Recognized testing agency or laboratory. The testing agency or laboratory should be one which is regularly engaged in the examination, testing, and evaluation of fire hose, which has an established factory inspection, listing, and labeling program; and which has published standards for listing and labeling that are nationally recognized. For example, the Underwriters' Laboratories, Inc., 333 Pfingstein Road, Northbrook, IL 60062 and the Factory Mutual Engineering Division, 1151 Boston-Providence Turnpike, Norwood, MA 02062 are considered recognized testing agencies.

6.4.1 It should be noted that all applicable requirements of this specification should be verified in accordance with section 4. However, hose bearing the label of a recognized testing agency or laboratory will require additional testing only for those requirements, as applicable, which have not been performed or confirmed by the agency label service.

6.5 Information for the Norton standard abrasion wheel specified in 4.6.8.2 may be obtained from the Norton Co., 1 New Bond, Worchester, MA 01606.

6.6 First article. When a first article inspection is required, the unit should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of one hose assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4210-N191)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side) FIGHTING AND OTHER WATER SERVIC

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24606A(SH)	DOUBLE SYNTHETIC JACKET, WITH COUPLINGS, FIRE HOSE ASSEMBLIES, CHLOROSULFONATED POLYETHYLENE IMPREGNATED,

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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